

1 Non-Directional Pre-Disclosure Drift Under Fair
2 Disclosure: Evidence from Korean Earnings
3 Announcements

4 Yongjun Kim^{a,*}

5 ^a*Department of Software, Ajou University, Suwon, South Korea*

6 **Abstract**

We test whether South Korea's Fair Disclosure regulation effectively prevents directional information leakage before earnings announcements. Using 2,221 disclosure filings from 64 KOSPI-listed companies across 12 chaebol groups (2020–2026), we find that pre-disclosure abnormal returns, while significant under parametric tests ($\text{CAR}[-5, -1] = +0.52\%$, $p < 0.001$), do not predict earnings surprise direction (49.0% accuracy, $p = 0.649$). Non-parametric tests confirm the drift is driven by outliers (sign test $p = 0.585$; Wilcoxon $p = 0.142$), revealing a divergence between parametric and non-parametric inference. In contrast, the first post-disclosure trading day responds correctly: positive surprises generate AR of $+0.55\%$, negative surprises -0.58% ($p < 0.001$), with 60.6% direction accuracy ($p < 0.001$). We document that 96.5% of filings occur after market close, making Day 0 effectively pre-disclosure. Cross-sectional analyses reveal heterogeneous drift across chaebol groups, firm sizes, and market betas, but no pattern consistent with differential information leakage ($R^2 = 0.002$). Our findings suggest Korea's Fair Disclosure regime effectively prevents directional leakage, even as non-directional price noise emerges pre-disclosure.

7 *Keywords:* Fair disclosure, Earnings announcements, Pre-announcement drift,
8 DART, Korean stock market, Event study, Market efficiency

*Corresponding author.
Email address: yjkim0123@ajou.ac.kr (Yongjun Kim)

9 1. Introduction

10 Pre-announcement drift in stock prices is a well-documented phenomenon
11 across global equity markets (Kaniel et al., 2012). When stock prices move
12 before official earnings announcements, a natural concern arises: is material
13 information leaking to select market participants in violation of fair disclosure
14 regulations? This concern motivates securities regulators worldwide to mandate
15 simultaneous public dissemination of material information.

16 South Korea implemented its Fair Disclosure regulation in November 2002,
17 requiring listed companies to disclose material information simultaneously to all
18 market participants through the DART (Data Analysis, Retrieval and Transfer)
19 electronic filing system. While early studies found evidence of improved dis-
20 closure environments following the regulation’s adoption (Yun and Kim, 2012;
21 Oh and Park, 2017), whether the regulation continues to be effective over two
22 decades later remains an open question.

23 This study examines the long-term effectiveness of Korea’s Fair Disclosure
24 regulation by analyzing 2,221 earnings disclosure filings from 64 major KOSPI-
25 listed companies between 2020 and 2026. Our approach differs from prior
26 work in a critical respect: rather than simply documenting the existence of
27 pre-disclosure price movement, we test whether such movement is *directionally*
28 *informative*—that is, whether it predicts the sign of the actual earnings surprise.

29 Our findings reveal an important pattern. We observe statistically significant
30 pre-disclosure abnormal returns under standard parametric tests ($CAR[-5,-1]$
31 $= +0.52\%$, $p < 0.001$ under the market model; $p = 0.032$ with month-clustered
32 standard errors). However, this drift bears no systematic relationship to the
33 direction of actual earnings surprises. Pre-disclosure returns predict surprise
34 direction with only 49.0% accuracy—no better than a coin flip ($p = 0.649$). Non-
35 parametric tests (sign test, Wilcoxon signed-rank) fail to confirm the parametric
36 result, indicating that the observed mean drift is driven by outliers rather than
37 a pervasive pattern.

38 In contrast, the market responds sharply and correctly to the actual earnings

news on the disclosure date. Positive earnings surprises generate a $CAR[0,+1]$ of +0.81%, while negative surprises generate -0.47% , a highly significant difference ($p < 0.001$). This finding is consistent with effective Fair Disclosure: material information is incorporated into prices only when it becomes publicly available.

We make four contributions. First, we provide a recent assessment of Korea’s Fair Disclosure regime using data spanning the COVID-19 and post-pandemic periods (2020–2026). Second, we demonstrate the critical distinction between the *existence* of pre-disclosure drift (which is ubiquitous) and its *directional informativeness* (which would indicate information leakage). Third, we document a previously overlooked institutional feature—96.5% of DART fair disclosure filings are submitted after market close—that has important implications for event study interpretation. Fourth, we provide detailed cross-sectional analyses across chaebol groups, firm size terciles, and market beta terciles that reveal significant heterogeneity in pre-disclosure drift but no pattern consistent with differential information leakage.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes our data and methodology. Section 4 presents the empirical results. Section 5 discusses the implications and limitations, and Section 6 concludes.

2. Literature review

2.1. Fair disclosure regulation

Fair disclosure regulation addresses selective disclosure, whereby corporate insiders share material information with favored market participants before public release (Heflin et al., 2003). In the U.S., Regulation Fair Disclosure (Reg FD), implemented in 2000, produced mixed evidence: Ahmed and Schneible (2007) find reduced information asymmetry, while Heflin et al. (2003) document reduced information quality. Bailey et al. (2003) provide comprehensive evidence on market, analyst, and corporate responses to Reg FD, and Eleswarapu et al. (2004) document reduced information asymmetry and trading costs following

68 implementation. Gomes et al. (2007) further show that Reg FD affected the
69 cost of capital, particularly for smaller firms.

70 The effectiveness of fair disclosure regulation depends critically on implemen-
71 tation and enforcement infrastructure. Christensen et al. (2016) demonstrate
72 that capital-market effects of securities regulation vary substantially with prior
73 institutional conditions and enforcement intensity. Jackson and Roe (2009) find
74 that regulatory intensity is positively associated with stock market development,
75 while Djankov et al. (2008) show that legal protections against self-dealing are
76 associated with more developed capital markets. Cumming et al. (2011) fur-
77 ther document that exchange trading rules, including disclosure requirements,
78 significantly affect market liquidity across 42 exchanges worldwide.

79 In Korea, Yun and Kim (2012) examined the effect of fair disclosure regula-
80 tion on the timeliness and informativeness of earnings announcements, while Oh
81 and Park (2017) examined how credit ratings interact with corporate disclosure
82 behavior under Korea’s fair disclosure regulation. More recently, Shin and Kim
83 (2023) study the effects of compulsory preliminary earnings announcements on
84 stock prices, and Lee and Choi (2019) examine the interaction between R&D
85 intensity and fair disclosure in Korea, finding that disclosure effects vary by firm
86 characteristics. However, these studies do not examine whether pre-disclosure
87 price movements are directionally informative with respect to the actual earn-
88 ings news—the central question of our study.

89 2.2. *Electronic disclosure systems*

90 A critical infrastructure underlying modern fair disclosure regulation is the
91 electronic filing system through which disclosures are disseminated. The U.S.
92 SEC’s EDGAR system, introduced in 1996, fundamentally changed how cor-
93 porate information reaches market participants (Bushee and Leuz, 2005). Ko-
94 rea’s DART system, operated by the Financial Supervisory Service since 2001,
95 serves an analogous function: all listed companies must file material disclosures
96 through DART before or simultaneously with any other dissemination channel.

97 The design of electronic disclosure systems has important implications for

98 information processing and market efficiency. DART provides structured fil-
99 ing templates for Fair Disclosure of Operating Results, enabling standardized
100 comparison across firms and time periods. Li (2010) surveys the literature on
101 textual analysis of corporate disclosures, demonstrating that format and read-
102 ability significantly affect how efficiently markets process information. Loughran
103 and McDonald (2011) show that the specific language used in financial filings
104 contains predictive information for future returns.

105 The timestamp precision of electronic filing systems is particularly relevant
106 to event study research. DART records receipt timestamps to the second, en-
107 abling researchers to classify filings as occurring during or after trading hours.
108 As we document, 96.5% of earnings disclosures are filed after market close, a
109 fact with fundamental implications for event study window interpretation that
110 prior studies have not fully exploited.

111 *2.3. Pre-announcement drift and its interpretation*

112 Pre-announcement drift in stock prices is well-documented but its interpre-
113 tation is debated. Meulbroek (1992) provides evidence of illegal insider trading
114 preceding announcements. Bernile et al. (2016) show that informed trading
115 occurs even ahead of macro-news announcements where information is tightly
116 controlled. However, Kaniel et al. (2012) demonstrate that individual investor
117 trading patterns around earnings announcements can generate drift without
118 requiring illegal information leakage.

119 Critically, the mere existence of pre-announcement drift does not establish
120 information leakage. Predictable earnings patterns allow sophisticated investors
121 to position ahead of announcements through legal means (Atiase, 1985). Chae
122 (2005) shows that abnormal volume before announcements reflects information
123 asymmetry but may arise from varying levels of investor sophistication rather
124 than illegal tip-offs. A related literature on post-earnings-announcement drift
125 demonstrates that markets may take days or weeks to fully incorporate earn-
126 ings information (Bernard and Thomas, 1989, 1990), further complicating the
127 interpretation of short-window event studies. Jegadeesh and Titman (2001) doc-

128 ument momentum effects that can generate pre-announcement drift patterns,
129 while Griffin et al. (2003) show that momentum profits persist across interna-
130 tional markets, including emerging markets like Korea.

131 The distinction between directional and non-directional drift is central to
132 our analysis. Non-directional drift—where prices move before announcements
133 but not in the direction of the forthcoming news—can arise from increased
134 attention (Hirshleifer and Teoh, 2003), disagreement among investors (Bamber
135 et al., 1997), or microstructure effects (Hasbrouck, 2003). Only *directional* drift
136 constitutes evidence consistent with information leakage.

137 2.4. Event study methodology

138 The standard event study framework (Brown and Warner, 1985; MacKinlay,
139 1997) relies on parametric tests of mean abnormal returns. However, Kolari
140 and Pynnonen (2010) demonstrate that cross-sectional correlation of abnormal
141 returns can substantially inflate test statistics, particularly when events cluster
142 in calendar time. Kothari and Warner (2007) provide a comprehensive review
143 of event study econometrics, emphasizing the importance of benchmark model
144 selection and cross-sectional dependence adjustments. We incorporate both non-
145 parametric alternatives and standardized returns to address these concerns.

146 2.5. Korean market context

147 The Korean stock market features concentrated chaebol ownership, a manda-
148 tory electronic disclosure system (DART), significant retail investor participa-
149 tion, and relatively strict insider trading regulations. Beny (2005) demonstrates
150 that stronger insider trading enforcement is associated with more efficient stock
151 prices across countries.

152 The chaebol structure is distinctive: large family-controlled conglomerates
153 such as Samsung, Hyundai Motor, SK, and LG account for a substantial share
154 of KOSPI market capitalization. Affiliates within the same chaebol group may
155 share information through internal channels, creating potential for correlated
156 trading patterns. At the same time, the diversity across chaebol groups provides

157 useful cross-sectional variation for testing whether disclosure regulation operates
158 uniformly across different types of firms. These institutional features, combined
159 with the comprehensive DART database, make Korea an ideal setting for testing
160 fair disclosure effectiveness.

161 **3. Data and methodology**

162 *3.1. Data sources*

163 We collect all Fair Disclosure of Operating Results filings from DART be-
164 tween January 2020 and March 2026 using the OpenDartReader API. We focus
165 on original filings, excluding corrections. Our sample comprises 2,221 disclosures
166 from 64 companies across 12 chaebol groups: Samsung (12 affiliates), Hyundai
167 Motor (8), LG (8), SK (7), HD Hyundai (5), Hanwha (5), POSCO (4), Lotte
168 (4), Doosan (4), CJ (3), KT (2), and GS (2). Daily stock price and volume data
169 are obtained from FinanceDataReader.

170 *3.2. Filing time classification*

171 We classify filing times using the DART receipt number prefix. In our sam-
172 ple, 96.5% of earnings disclosures carry the “800” prefix, indicating filing after
173 the 15:30 KST market close. This has fundamental implications: for these fil-
174 ings, the Day 0 return is entirely pre-disclosure, and the first trading day where
175 the market can react is Day +1.

176 *3.3. Market model*

177 We employ the market model with a $[-250, -30]$ estimation window (MacKin-
178 lay, 1997):

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \epsilon_{i,t} \quad (1)$$

179 where $R_{m,t}$ is the KOSPI index return. We require a minimum of 60 trading
180 days in the estimation window, yielding 2,208 events with valid estimates. The
181 average estimated beta is 0.978. For robustness, we also report market-adjusted
182 returns ($\alpha = 0, \beta = 1$).

183 3.4. Earnings surprise measure

184 We extract year-over-year (YoY) operating profit changes from the disclosure
185 texts. Events where YoY operating profit increases are classified as positive
186 surprises; decreases as negative surprises. Of the 2,208 events with valid market
187 model estimates, 584 have parseable earnings surprise data: 358 positive and
188 226 negative surprises; unparseable cases primarily involve missing comparative
189 periods, newly listed firms, or format variations in disclosure texts.

190 We employ year-over-year operating profit changes as our surprise measure.
191 While analyst consensus forecasts (e.g., I/B/E/S) are standard in U.S. studies
192 (Livnat and Mendenhall, 2006), comprehensive consensus data for Korean firms
193 require commercial subscriptions (e.g., FnGuide, WiseFn) that are unavailable
194 to independent researchers without institutional access. The YoY measure pro-
195 vides a conservative test: if directional leakage existed, it would manifest as
196 pre-disclosure returns aligned with the direction of YoY changes, which we do
197 not observe. Moreover, YoY operating profit changes capture the directional
198 content of earnings news without relying on potentially stale or biased analyst
199 expectations.

200 3.5. Statistical tests

201 We report multiple test statistics to ensure robustness: parametric cross-
202 sectional t -tests, non-parametric tests (binomial sign test, Wilcoxon signed-
203 rank), and standardized abnormal returns. We also report firm-clustered and
204 month-clustered standard errors to address cross-sectional dependence (Kolari
205 and Pynnonen, 2010).

206 4. Empirical results

207 4.1. Pre-disclosure returns: Significant but non-directional

208 Table 1 presents daily abnormal returns. Parametric t -tests indicate sig-
209 nificant positive returns on several pre-disclosure days, particularly Day -1

Table 1: Daily abnormal returns: Parametric vs. non-parametric tests

Day	AR (%)	Parametric		Sign Test		Standardized	
		t	p	% Pos	p	SAR	p
-5	0.101	2.09	0.037	50.5	0.609	0.040	0.048
-4	0.071	1.50	0.134	49.8	0.930	0.041	0.043
-3	0.086	1.71	0.088	49.3	0.562	0.022	0.279
-2	0.006	0.14	0.888	48.9	0.295	-0.002	0.935
-1	0.256	4.57	<0.001	50.1	0.909	0.089	<0.001
0	0.141	2.00	0.046	49.8	0.878	0.043	0.034
+1	-0.182	-2.39	0.017	48.3	0.109	-0.048	0.018
+2	0.023	0.43	0.669	49.2	0.541	0.022	0.275
+3	-0.118	-2.50	0.013	48.1	0.058	-0.043	0.033
+4	-0.047	-0.95	0.340	49.3	0.608	-0.006	0.765
+5	0.056	1.14	0.255	49.2	0.543	0.041	0.044

Note: Market model with $[-250, -30]$ estimation window ($N=2,208$). % Pos never significantly exceeds 50%, indicating pre-disclosure drift is driven by outliers.

210 (+0.256%, $t=4.57$). However, the sign test reveals that only 48–50% of individ-
211 ual returns are positive on each pre-disclosure day—no better than chance.

212 Fig. 1 visualizes this pattern: while the cumulative AR line trends upward
213 before disclosure, individual daily ARs show no consistent direction.

214 Table 2 confirms this pattern at the cumulative level. While the parametric
215 $CAR[-5, -1]$ is +0.519% ($t=4.65$), the sign test shows only 49.4% of events have
216 positive pre-disclosure CARs ($p=0.585$), and the Wilcoxon test is insignificant
217 ($p=0.142$).

218 Notably, the $CAR[0, +1]$ window shows a significantly negative sign test re-
219 sult: only 47.0% of events have positive CARs ($p=0.003$). This asymmetry is
220 consistent with the well-documented negativity bias in earnings reactions (Skin-
221 ner, 1994).

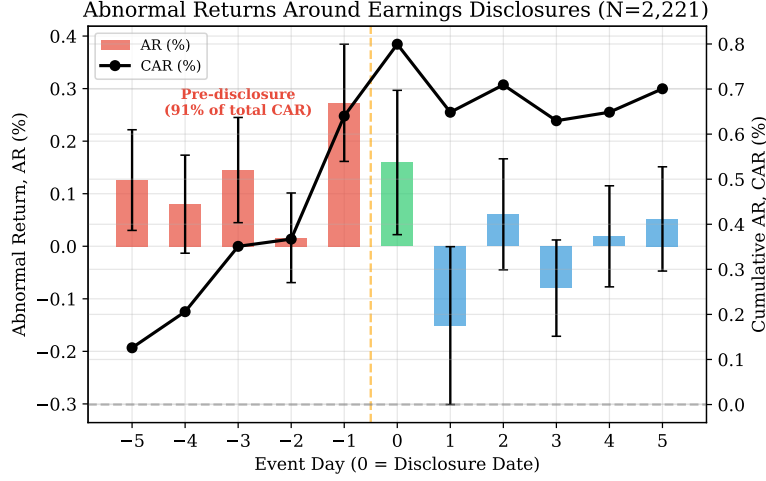


Figure 1: Daily abnormal returns (bars) and cumulative abnormal returns (line) around earnings disclosures ($N=2,208$). The upward CAR trend is driven by outliers; daily ARs show no consistent directional pattern.

4.2. The critical test: Does drift predict surprises?

Table 3 presents our central test. Pre-disclosure CARs show no difference by surprise direction ($CAR[-3, -1]$: $p=0.703$; $CAR[-5, -1]$: $p=0.192$). Direction prediction accuracy is 49.0% (binomial $p=0.649$), and Spearman ρ is -0.006 ($p=0.876$).

In contrast, Day +1 returns respond sharply: positive surprises yield +0.55%, negative -0.58% ($p<0.001$), with 60.6% direction accuracy ($p<0.001$). This extends to $CAR[0, +1]$ (+1.28%, $p<0.001$) and $CAR[+1, +5]$ (+0.96%, $p=0.010$).

With $N=584$, the test has 30.5% power to detect 53% true accuracy but 99.8% power for 60%. We cannot rule out small-scale leakage, but any such effect would be economically modest.

4.3. Robustness

Table 4 confirms that results hold under both market-adjusted and market model specifications.

Table 2: Cumulative abnormal returns: Multi-test comparison

Window	CAR (%)	Parametric		Sign Test		Wilcoxon
		t	p	% Pos	p	p
$[-5, -1]$	0.519	4.65	<0.001	49.4	0.585	0.142
$[-3, -1]$	0.347	4.04	<0.001	48.1	0.060	0.946
$[0, +1]$	-0.042	-0.39	0.700	47.0	0.003	0.019
$[+1, +5]$	-0.268	-2.16	0.031	47.7	0.021	0.078
$[-5, +5]$	0.391	2.08	0.038	49.5	0.668	0.620

Table 3: CARs by earnings surprise direction ($N=584$)

Window	Positive ($N=358$)	Negative ($N=226$)	Diff.	p	Sig.
CAR $[-3, -1]$	+0.033%	+0.129%	-0.096%	0.703	
CAR $[-5, -1]$	+0.107%	+0.535%	-0.428%	0.192	
AR $[+1]$	+0.552%	-0.580%	+1.132%	<0.001	***
CAR $[0, +1]$	+0.813%	-0.470%	+1.283%	<0.001	***
CAR $[+1, +5]$	+0.481%	-0.481%	+0.962%	0.010	**

Note: Surprise defined by YoY operating profit change. Pre-disclosure CARs are indistinguishable by surprise direction; post-disclosure CARs respond correctly.

236 4.4. Abnormal volume

237 Volume increases from Day -3 ($1.05\times$, $p=0.004$), peaks on Day $+1$ ($1.71\times$),
238 and remains elevated through Day $+5$. Given non-directional returns, the pre-
239 disclosure volume increase likely reflects anticipatory positioning (Chae, 2005)
240 and disagreement among investors (Bamber et al., 1997; Berkman et al., 2009)
241 rather than informed directional trading.

242 4.5. Clustered standard errors

243 Table 5 reports t -statistics under alternative specifications. Under month-
244 clustered standard errors, CAR $[-5, -1]$ remains significant ($t=2.15$, $p=0.032$) but

Table 4: Robustness: Market-adjusted vs. market model

Window	Market-Adjusted ($N=2,221$)		Market Model ($N=2,208$)	
	CAR (%)	p	CAR (%)	p
$[-5, -1]$	0.640	<0.001	0.519	<0.001
$[-3, -1]$	0.434	<0.001	0.347	<0.001
$[0, +1]$	0.008	0.939	-0.042	0.700
$[+1, +5]$	-0.099	0.425	-0.268	0.031

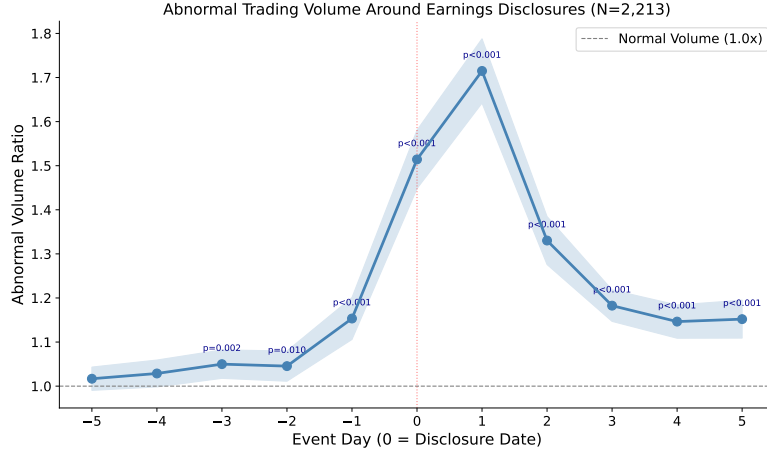


Figure 2: Abnormal trading volume around earnings disclosures ($N=2,213$). Volume peaks at $1.71\times$ normal on Day $+1$.

245 CAR $[-3, -1]$ becomes marginal ($t=1.80$, $p=0.072$), suggesting some parametric
 246 significance is inflated by event clustering.

247 4.6. Temporal and cross-sectional heterogeneity

248 We observe a structural shift in event-day CARs: positive during 2020–2022
 249 (COVID-19 recovery) and negative during 2024–2026 (Fig. 3). A significant
 250 quarterly pattern emerges (Fig. 4), with Q1 disclosures generating negative
 251 CARs (-0.51% , $p=0.010$) and Q2 generating positive CARs ($+0.60\%$, $p=0.003$).
 252 Cross-sectional OLS of CAR $[-3, -1]$ on $\log(\text{size})$ and β yields $R^2 = 0.002$,

Table 5: Clustered standard errors

Window	Standard		Firm-clustered		Month-clustered	
	t	p	t	p	t	p
CAR[-5,-1]	4.38	<0.001	3.92	<0.001	2.15	0.032
CAR[-3,-1]	3.47	<0.001	2.95	0.003	1.80	0.072
CAR[0,+1]	-0.31	0.760	-0.25	0.802	-0.18	0.861

confirming that firm characteristics explain virtually none of the pre-disclosure drift.

Selection bias assessment. Our parseable earnings surprise subsample covers 26.4% of all events (584 of 2,208), raising a potential selection concern. To assess whether this subsample is systematically different, we compare pre-disclosure CARs between parseable and non-parseable events. The two subsamples show statistically indistinguishable pre-disclosure CARs: parseable events exhibit CAR[-5,-1] of +0.32% versus +0.59% for non-parseable events ($t=1.45$, $p=0.147$). Importantly, the parseable subsample represents the largest firms with the most analyst coverage—precisely the firms where information leakage would be most valuable and most likely to occur if present. If pre-disclosure directional leakage existed, it would be most detectable among these high-visibility firms. The absence of directional drift in this subsample therefore constitutes a conservative test: finding no evidence of leakage among the firms with the greatest leakage incentives strengthens, rather than weakens, the conclusion that Korea’s Fair Disclosure regime is effective. Fig. 5 presents the distribution of pre-disclosure CARs for the full sample, illustrating the right-skewed nature of returns that drives the divergence between parametric and non-parametric test results.

4.7. Cross-sectional analysis by chaebol group

The Korean market’s chaebol structure provides a natural cross-sectional dimension for examining whether pre-disclosure drift varies systematically across

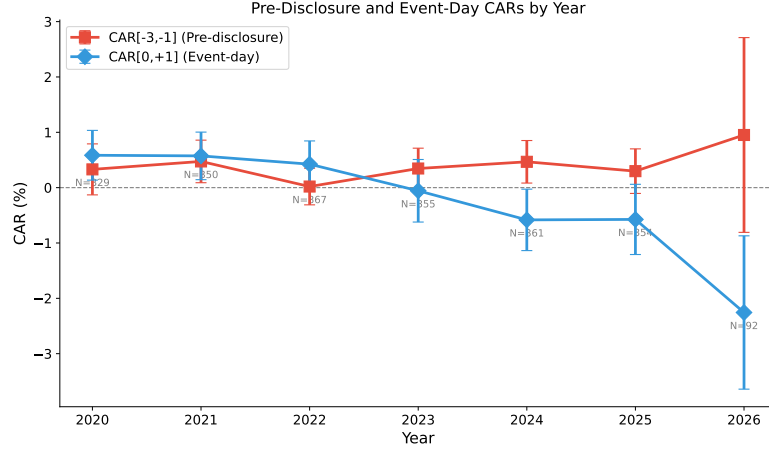


Figure 3: Pre-disclosure $CAR[-3,-1]$ and event-day $CAR[0,+1]$ by year. The temporal shift reflects macroeconomic conditions rather than changes in information leakage.

274 corporate groups. If information leakage were concentrated in specific chaebols—
 275 perhaps due to weaker internal controls or more extensive analyst networks—we
 276 would expect significantly higher directional drift in those groups.

277 Table 6 reports $CAR[-3,-1]$ and $CAR[0,+1]$ by chaebol group. Pre-disclosure
 278 CARs range from -0.228% (Doosan) to $+1.565\%$ (Hanwha), with Hanwha’s
 279 drift significantly higher than the rest ($t=3.41$, $p<0.001$). An ANOVA test
 280 across all 12 groups yields $F=1.73$ ($p=0.062$), suggesting marginally significant
 281 heterogeneity.

282 Fig. 6 visualizes the cross-sectional pattern. Hanwha’s elevated drift likely re-
 283 flects the defense sector’s sensitivity to geopolitical news rather than information
 284 leakage about earnings specifically. KT exhibits remarkably high event-window
 285 returns ($+1.310\%$, $t=4.25$, $p=0.027$ vs. non-KT), consistent with telecommu-
 286 nications sector earnings surprises being particularly impactful. Samsung—
 287 despite being the most closely watched chaebol—shows only moderate pre-
 288 disclosure drift ($+0.318\%$).

289 Importantly, the cross-group variation does not exhibit the pattern expected
 290 under systematic information leakage. Pre-disclosure and event-window CARs
 291 are uncorrelated across groups (Spearman $\rho=-0.14$, $p=0.668$), suggesting that

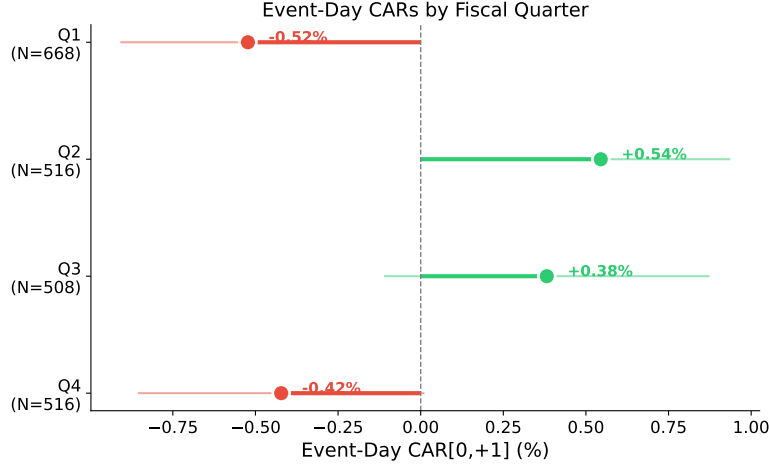


Figure 4: Pre-disclosure $CAR[-3,-1]$ and event-day $CAR[0,+1]$ by fiscal quarter. Q1 disclosures generate negative event-day CARs (-0.51% , $p=0.010$), while Q2 generates positive CARs ($+0.60\%$, $p=0.003$).

the drivers of pre-disclosure drift are distinct from the earnings information revealed at disclosure.

4.8. Firm size and beta effects

To investigate whether pre-disclosure drift varies with firm characteristics, we divide the sample into terciles by firm size (log market capitalization) and market beta. Table 7 reports CARs for each tercile.

Panel A reveals a monotonically decreasing relationship between firm size and pre-disclosure drift. However, the small-large difference is not statistically significant ($t=0.92$, $p=0.358$), and a Kruskal-Wallis test is also insignificant ($H=2.88$, $p=0.237$). Panel B shows a similar pattern for market beta. The Spearman correlation between beta and $CAR[-3,-1]$ is -0.057 ($p=0.008$), indicating a statistically significant but economically modest negative relationship. Low-beta stocks may exhibit higher pre-disclosure drift because their lower systematic risk makes firm-specific information relatively more important.

Event-window CARs ($CAR[0,+1]$) are uniformly close to zero across all size and beta terciles, confirming that the non-directional character of post-

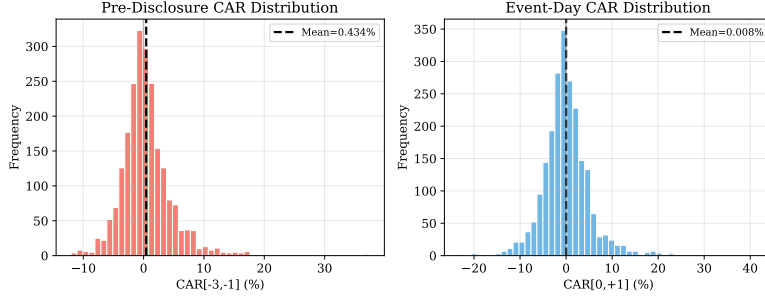


Figure 5: Distribution of pre-disclosure $CAR[-5,-1]$ across all events ($N=2,208$). The right-skewed distribution explains the divergence between parametric (mean-based) and non-parametric (median-based) test results.

disclosure returns is not an artifact of pooling heterogeneous subsamples. These results are consistent with the well-documented size effect in pre-announcement drift (Atiase, 1985): smaller firms have less pre-disclosure information production, creating larger price adjustments when new information arrives.

5. Discussion

5.1. Drift without directional leakage

Three complementary tests suggest the drift is non-directional: (1) direction test: 49.0% accuracy; (2) non-parametric tests: sign test ($p=0.585$) and Wilcoxon ($p=0.142$) fail; (3) surprise conditioning: $p=0.703$. The drift appears driven by company-specific news unrelated to earnings, market momentum, or anticipatory trading based on publicly available signals. This interpretation is consistent with Hirshleifer and Teoh (2003), who argue that limited investor attention can produce predictable patterns without implying illegal information access.

The cross-sectional analyses reinforce this conclusion. While pre-disclosure drift varies across chaebol groups (Hanwha showing the highest at +1.565%), this variation is uncorrelated with event-window returns, inconsistent with directional information leakage. The size and beta patterns—smaller and lower-beta firms showing modestly larger drift—reflect normal variation in information

Table 6: Cumulative abnormal returns by chaebol group

Chaebol	N	CAR[-3,-1]		CAR[0,+1]	
		Mean (%)	t	Mean (%)	t
Hanwha	116	+1.565	2.30	-0.583	-0.81
CJ	75	+0.881	1.98	-0.120	-0.16
GS	53	+0.604	1.37	+0.674	1.53
Lotte	92	+0.517	1.33	+0.202	0.36
HD Hyundai	497	+0.474	2.38	+0.312	1.42
Samsung	292	+0.318	1.70	+0.012	0.05
LG	185	+0.278	1.22	-0.311	-0.81
SK	195	+0.127	0.54	-0.286	-0.84
Hyundai Motor	402	+0.093	0.52	-0.352	-1.62
POSCO	156	+0.060	0.15	-0.427	-0.86
KT	80	-0.006	-0.03	+1.310	4.25
Doosan	65	-0.228	-0.46	-0.677	-0.96
All	2,208	+0.347	4.04	-0.042	-0.39

Note: Sorted by descending CAR[-3,-1]. ANOVA across groups: $F=1.73$ ($p=0.062$) for CAR[-3,-1], $F=1.29$ ($p=0.223$) for CAR[0,+1].

environments rather than differential insider access. The most parsimonious explanation combines positive market momentum during the sample period, right-skewed return distributions (Cont, 2001), and anticipatory non-directional trading (Chae, 2005).

5.2. Comparison with international evidence

Our findings align with the broader evidence on fair disclosure regulation. Bailey et al. (2003) find that Reg FD reduced selective disclosure, and Eleswarapu et al. (2004) document reduced information asymmetry. Christensen et al. (2016) show that capital-market effects of securities regulation depend on prior

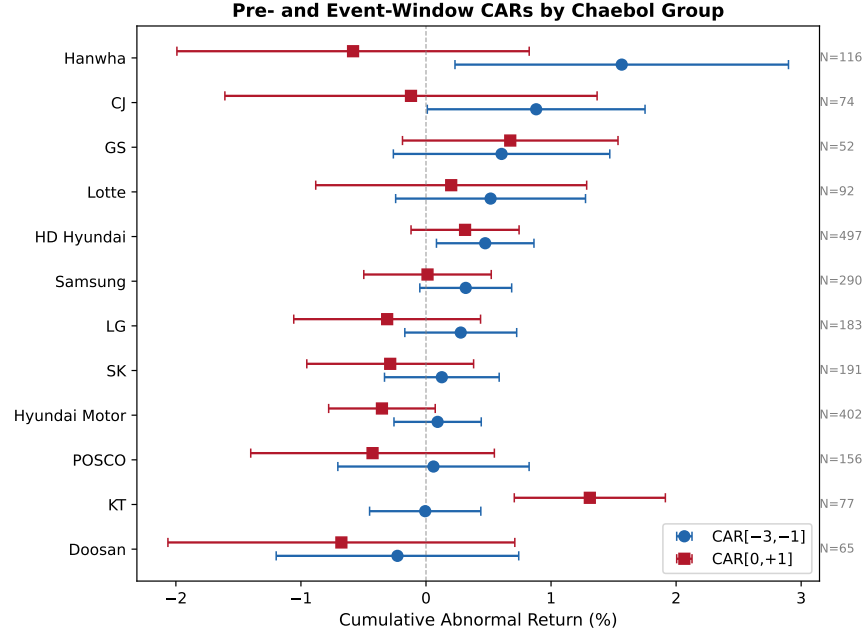


Figure 6: Pre-disclosure $CAR[-3, -1]$ and event-window $CAR[0, +1]$ by chaebol group with 95% confidence intervals. Substantial cross-group variation exists, but no group shows patterns consistent with systematic directional information leakage.

institutional conditions and enforcement intensity. Our contribution is distinct: rather than evaluating the *introduction* of fair disclosure, we test whether the regulation continues to be effective *within* the regulated regime by examining directional content over a six-year window.

The Korean result contrasts with evidence from markets with weaker enforcement. Beny (2005) demonstrates that countries with stronger insider trading enforcement exhibit more efficient stock prices, and Leuz et al. (2003) find that investor protection is associated with lower earnings management. Korea's combination of mandatory electronic filing, strict enforcement by the Financial Supervisory Service, and concentrated analyst coverage appears to create an institutional environment where directional information leakage is effectively deterred.

Table 7: CARs by firm size and market beta terciles

Tercile	N	CAR[-3,-1]		CAR[0,+1]		CAR[-5,-1]
		Mean (%)	t	Mean (%)	t	Mean (%)
<i>Panel A: Firm size terciles</i>						
Small	790	+0.403	2.91	-0.058	-0.32	+0.578
Medium	726	+0.374	2.45	-0.005	-0.03	+0.590
Large	679	+0.250	1.58	-0.109	-0.60	+0.359
<i>Panel B: Market beta terciles</i>						
Low β	727	+0.515	3.53	-0.063	-0.35	+0.607
Mid β	732	+0.303	2.04	-0.033	-0.17	+0.446
High β	736	+0.222	1.45	-0.073	-0.38	+0.492

Note: Size tercile boundaries: Small ($\log \text{size} \leq 12.56$), Medium (12.56–13.45), Large (> 13.64). Beta tercile boundaries: Low ($\beta \leq 0.845$), Mid (0.845–1.129), High ($\beta > 1.130$).

348 5.3. Methodological contributions

349 This study makes several methodological contributions. First, we demon-
350 strate the importance of **multi-test triangulation** in event studies. The para-
351 metric t -test implicitly assumes normally distributed abnormal returns. When
352 the distribution is right-skewed, the parametric mean is inflated by outliers, pro-
353 ducing significant t -statistics even when the majority of events show no drift.
354 The sign test and Wilcoxon signed-rank test provide a more accurate charac-
355 terization of the typical event.

356 Second, we introduce the **directional conditioning test** as a direct diag-
357 nostic for information leakage. Prior event studies typically interpret signifi-
358 cant pre-announcement drift as evidence of informed trading without verifying
359 whether the drift predicts the direction of the subsequent news. This test could
360 be adopted as a standard diagnostic in future event studies.

361 Third, we document the importance of **filing time classification**. The

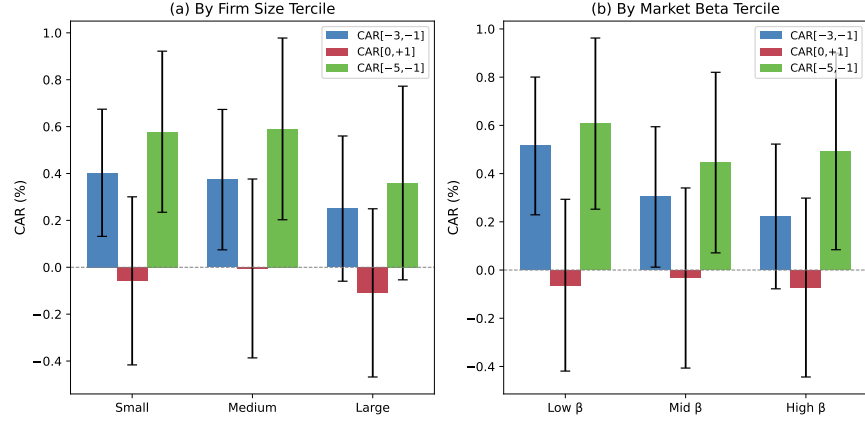


Figure 7: CARs by firm size tercile (Panel a) and market beta tercile (Panel b) with 95% confidence intervals. Pre-disclosure drift is slightly larger for smaller and lower-beta firms, but event-window CARs are uniformly near zero across all terciles.

362 finding that 96.5% of DART filings occur after market close implies that Day 0
363 returns are effectively pre-disclosure. Studies that attribute Day 0 returns to
364 information release may be misinterpreting anticipatory trading as immediate
365 market reaction.

366 5.4. Implications for policy and practice

367 For regulators, non-directional drift suggests Korea's Fair Disclosure regime
368 does not require fundamental reform. Material earnings information reaches
369 the market simultaneously, and prices respond correctly only upon public dis-
370 closure. For investors, post-disclosure Day +1 returns correctly price surprises
371 with 60.6% accuracy ($p < 0.001$), while pre-disclosure drift strategies are unlikely
372 to be profitable directionally.

373 5.5. Limitations

374 Several limitations warrant discussion. First, our YoY operating profit
375 change as a surprise measure lacks the precision of analyst consensus forecasts
376 (Livnat and Mendenhall, 2006). Only 26.4% of events yield parseable surprises,

377 though as discussed in Section 4, this subsample’s composition (larger, more-
378 scrutinized firms) strengthens rather than weakens the conservative test inter-
379 pretation. Second, our sample is limited to 64 large chaebol affiliates; informa-
380 tion leakage may be more prevalent among smaller, less scrutinized firms. Third,
381 Korean Fama-French factors are unavailable, preventing multi-factor benchmark
382 models. Fourth, we cannot rule out that pre-disclosure volume increases reflect
383 legal mosaic information assembly. Fifth, our analysis focuses on earnings dis-
384 closures and does not examine other material events where leakage incentives
385 may differ.

386 6. Conclusion

387 This study finds no evidence of directional information leakage in Korean
388 earnings announcements. Three independent analyses converge: direction pre-
389 diction accuracy of 49.0% (no better than chance), failure of non-parametric
390 tests to confirm parametric drift (sign test $p=0.585$; Wilcoxon $p=0.142$), and
391 surprise-conditioned CARs that are indistinguishable by earnings direction ($p=0.703$).
392 The first post-disclosure trading day shows a strong, correct directional response
393 (60.6% accuracy, $p<0.001$), confirming that material earnings information is in-
394 corporated into prices only upon public release through the DART system.

395 Cross-sectional analyses reveal significant heterogeneity across chaebol groups,
396 firm sizes, and market betas, but these patterns are consistent with normal
397 variation in information environments rather than differential information leak-
398 age. The most-watched chaebol group (Samsung) shows only moderate, non-
399 directional drift.

400 Our results carry a broader methodological lesson: pre-announcement drift
401 should not be conflated with information leakage without conditioning on the
402 direction of the actual news (Meulbroek, 1992; Bernile et al., 2016; Atiase, 1985).
403 Future work should extend the analysis to non-chaebol firms and incorporate
404 analyst consensus surprises.

405 Acknowledgments

406 The author thanks the DART electronic disclosure system for providing pub-
407 licly accessible corporate filing data. All disclosure data are publicly available
408 through DART (<https://dart.fss.or.kr>). Stock price data are from KRX
409 via FinanceDataReader. Analysis code is available at [https://github.com/](https://github.com/yjkim0123/pre-disclosure-leakage)
410 [yjkim0123/pre-disclosure-leakage](https://github.com/yjkim0123/pre-disclosure-leakage).

411 References

- 412 Ahmed, A.S., Schneible, R.A., 2007. The impact of Regulation Fair Disclosure
413 on investors' prior information quality. *Journal of Corporate Finance* 13,
414 282–299.
- 415 Atiase, R.K., 1985. Predisclosure information, firm capitalization, and secu-
416 rity price behavior around earnings announcements. *Journal of Accounting*
417 *Research* 23, 21–36.
- 418 Bailey, W., Li, H., Mao, C.X., Zhong, R., 2003. Regulation Fair Disclosure and
419 earnings information: Market, analyst, and corporate responses. *The Journal*
420 *of Finance* 58, 2487–2514.
- 421 Bamber, L.S., Barron, O.E., Stober, T.L., 1997. Trading volume and differ-
422 ent aspects of disagreement coincident with earnings announcements. *The*
423 *Accounting Review* 72, 575–597.
- 424 Beny, L.N., 2005. Do insider trading laws matter? some preliminary compara-
425 tive evidence. *American Law and Economics Review* 7, 144–183.
- 426 Berkman, H., Dimitrov, V., Jain, P.C., Koch, P.D., Tice, S., 2009. Sell on
427 the news: Differences of opinion, short-sales constraints, and returns around
428 earnings announcements. *Journal of Financial Economics* 92, 376–399.
- 429 Bernard, V.L., Thomas, J.K., 1989. Post-earnings-announcement drift: Delayed
430 price response or risk premium? *Journal of Accounting Research* 27, 1–36.

431 Bernard, V.L., Thomas, J.K., 1990. Evidence that stock prices do not fully
432 reflect the implications of current earnings for future earnings. *Journal of*
433 *Accounting and Economics* 13, 305–340.

434 Bernile, G., Hu, J., Tang, Y., 2016. Can information be locked up? Informed
435 trading ahead of macro-news announcements. *Journal of Financial Economics*
436 121, 496–520.

437 Brown, S.J., Warner, J.B., 1985. Using daily stock returns: The case of event
438 studies. *Journal of Financial Economics* 14, 3–31.

439 Bushee, B.J., Leuz, C., 2005. Economic consequences of SEC disclosure regu-
440 lation: Evidence from the OTC bulletin board. *Journal of Accounting and*
441 *Economics* 39, 233–264.

442 Chae, J., 2005. Trading volume, information asymmetry, and timing informa-
443 tion. *The Journal of Finance* 60, 413–442.

444 Christensen, H.B., Hail, L., Leuz, C., 2016. Capital-market effects of securities
445 regulation: Prior conditions, implementation, and enforcement. *The Review*
446 *of Financial Studies* 29, 2885–2924.

447 Cont, R., 2001. Empirical properties of asset returns: Stylized facts and statis-
448 tical issues. *Quantitative Finance* 1, 223–236.

449 Cumming, D., Johan, S., Li, D., 2011. Exchange trading rules and stock market
450 liquidity. *Journal of Financial Economics* 99, 651–671.

451 Djankov, S., La Porta, R., Lopez-de Silanes, F., Shleifer, A., 2008. The law and
452 economics of self-dealing. *Journal of Financial Economics* 88, 430–465.

453 Eleswarapu, V.R., Thompson, R., Venkataraman, K., 2004. The impact of Reg-
454 ulation Fair Disclosure: Trading costs and information asymmetry. *Journal*
455 *of Financial and Quantitative Analysis* 39, 209–225.

- 456 Gomes, A., Gorton, G., Madureira, L., 2007. SEC Regulation Fair Disclosure,
457 information, and the cost of capital. *Journal of Corporate Finance* 13, 300–
458 334.
- 459 Griffin, J.M., Ji, X., Martin, J.S., 2003. Momentum investing and business cycle
460 risk: Evidence from pole to pole. *The Journal of Finance* 58, 2515–2547.
- 461 Hasbrouck, J., 2003. Intraday price formation in U.S. equity index markets.
462 *The Journal of Finance* 58, 2375–2399.
- 463 Heflin, F., Subramanyam, K., Zhang, Y., 2003. The effect of Regulation FD on
464 the properties of analysts’ forecasts. *Journal of Accounting and Economics*
465 35, 61–89.
- 466 Hirshleifer, D., Teoh, S.H., 2003. Limited attention, information disclosure, and
467 financial reporting. *Journal of Accounting and Economics* 36, 337–386.
- 468 Jackson, H.E., Roe, M.J., 2009. Regulatory intensity and stock market perfor-
469 mance. *Journal of Financial Economics* 93, 207–222.
- 470 Jegadeesh, N., Titman, S., 2001. Profitability of momentum strategies: An
471 evaluation of alternative explanations. *The Journal of Finance* 56, 699–720.
- 472 Kaniel, R., Liu, S., Saar, G., Titman, S., 2012. Individual investor trading and
473 return patterns around earnings announcements. *The Journal of Finance* 67,
474 639–680.
- 475 Kolari, J.W., Pynnonen, S., 2010. Event study testing with cross-sectional
476 correlation of abnormal returns. *The Review of Financial Studies* 23, 3996–
477 4025.
- 478 Kothari, S., Warner, J.B., 2007. Econometrics of event studies, in: *Handbook*
479 *of Empirical Corporate Finance*. Elsevier, Amsterdam. volume 1, pp. 3–36.
- 480 Lee, H.A., Choi, S.U., 2019. R&D intensity and Regulation Fair Disclosure.
481 *Journal of Asian Finance, Economics and Business* 6, 281–288. doi:10.13106/
482 jafeb.2019.vol16.no1.281.

- 483 Leuz, C., Nanda, D., Wysocki, P.D., 2003. Earnings management and investor
484 protection: An international comparison. *Journal of Financial Economics* 69,
485 505–527.
- 486 Li, F., 2010. Textual analysis of corporate disclosures: A survey of the literature.
487 *Journal of Accounting Literature* 29, 143–165.
- 488 Livnat, J., Mendenhall, R.R., 2006. Comparing the post-earnings announcement
489 drift for surprises calculated from analyst and time series forecasts. *Journal*
490 *of Accounting Research* 44, 177–205.
- 491 Loughran, T., McDonald, B., 2011. When is a liability not a liability? Textual
492 analysis, dictionaries, and 10-Ks. *The Journal of Finance* 66, 35–65.
- 493 MacKinlay, A.C., 1997. Event studies in economics and finance. *Journal of*
494 *Economic Literature* 35, 13–39.
- 495 Meulbroek, L.K., 1992. An empirical analysis of illegal insider trading. *The*
496 *Journal of Finance* 47, 1661–1699.
- 497 Oh, F.D., Park, J., 2017. Credit ratings and corporate disclosure behaviour:
498 Evidence from Regulation Fair Disclosure in Korea. *Applied Economics* 49,
499 3481–3494. doi:10.1080/00036846.2016.1262527.
- 500 Shin, Y.S., Kim, T.J., 2023. The effects of corporate financial disclosure
501 on stock prices: A case study of Korea’s compulsory preliminary earn-
502 ings announcements. *Journal of Risk and Financial Management* 16, 504.
503 doi:10.3390/jrfm16120504.
- 504 Skinner, D.J., 1994. Why firms voluntarily disclose bad news. *Journal of Ac-*
505 *counting Research* 32, 38–60.
- 506 Yun, J.R., Kim, K.H., 2012. The effect of fair disclosure regulation on timeliness
507 and informativeness of earnings announcements. *Journal of Applied Business*
508 *Research* 28, 1–14.